

**Predictors of Waist-to-Height Ratio Among U.S.
Adults: A Multiple Linear Regression Analysis Using
NHANES 2017–2018 Data**

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Background

National health and nutrition demographics help explore the factors influencing overall health. Understanding the leading factors in Health and Nutrition is consequential for identifying the underlying health trends within the sample. This comes from the National Health and Nutrition Examination Survey. This can yield actionable insights to inform specialized community efforts to address concerns within communities at greatest risk.

Much of the data collected for the National Health and Nutrition Examination Survey is valuable information about a person's physical health. There are a variety of valuable health points, but the literature suggests that the waist-height ratio is a strong outcome variable of interest. The literature suggests that a higher WHR is associated with greater visceral fat in the stomach and abdominal regions. WHR is defined as waist circumference divided by height and captures fat distribution rather than just weight. WHR is linked with outcomes such as cardiovascular disease, diabetes, and metabolic syndrome, all diseases that kill millions yearly. Obesity is a major public health problem in the U.S, and WHR varies genuinely across demographic groups like sex, race/ethnicity, and age. Understanding predictors could inform screening and intervention.

The NHANES is a nationally representative dataset of U.S. civilian adults. The data is collected through various methods, including surveys, physical health exams, and consultations. Numerous health data points are collected, but among the most important are smoking status and alcohol consumption habits, as well as uniquely identifiable information such as race, ethnicity, education level, and annual family income, which provide a clearer picture of the health trends emerging among U.S. civilian adults. NHANES is collected in standardized Mobile Examination Centers to ensure high-quality measurements.

This study uses multiple linear regression on the NHANES 2017-2018 dataset to identify the demographic and clinical factors most strongly associated with WHR in a nationally representative sample of adults. The goal is to inform targeted screening and public health intervention strategies within the communities identified.

Research Problem

This analysis investigated the following primary research question: which demographic, clinical, and lifestyle factors are significantly associated with the waist-to-height ratio in U.S. adults during the 2017 NHANES survey period? Secondary question includes: among significant risk factors, which modifiable factors, such as behavioral factors, after controlling for non-modifiable characteristics such as sex and race/ethnicity? And do socioeconomic indicators, including education and income, independently predict WHR beyond clinical variables? Together, these questions aim to distinguish between risk factors necessary for behavioral interventions to support targeted public health resource designation. Generally organized as follows:

- Among the risk factors, which ones best predict the waist-to-height ratio, for modifiable and non-modifiable risk factors?
- What behavior and lifestyle choices are associated with a higher WHR?
- Which race and sex predictors are most significant, and which communities should be flagged for intervention?

Model and Method

The analysis proceeded with the exploratory data analysis, variable selection, and model validation. The initial analysis examined the distributions of all continuous variables to assess model assumptions and identify any necessary transformations. BMI exhibited mild right skew (Shapiro-Wilk $W = 0.945$, $p < .001$); this motivated a log transformation prior to modeling. After this transformation, the $\log(\text{BMI})$ showed noticeable improvement in symmetry ($W = 0.994$), and this was used in place of raw BMI.

Stepwise regression using AIC in both directions was applied to the full model. This identified a large set of significant predictors. The procedure allows for removing variables at each step. The global F-test confirmed

that the final model explains significantly more variation in WHR than the null intercept-only model ($F = 3795$, $df = 14$, $p < 2.2e-16$), which leads to the rejection of the $H_0 : \beta_1 = \beta_2 = \dots = \beta_{14} = 0$. For the CAP variable, a nested F-test was used to evaluate the marginal contribution, failing to reject $H_0 : \beta_{CAP} = 0$ ($F = 1.17$, $p = 0.28$), validating its exclusion.

The sample consisted of about 4,382 civilian adults from the NHANES 2017-2018 cycle. The sample was balanced by Sex. Age ranged from 18 to 80 years, exploring a broad representation of the adult lifespan across demographic factors. Race/Ethnicity is not equally distributed, with NH-White participants comprising the largest group, followed by Hispanic, NH-Black, and NH-Other. This is an intentional oversampling of the minority population to ensure the statistical power for subgroup analysis.

Under the Gauss-Markov assumptions of linearity, exogeneity, homoscedasticity, and uncorrelated errors, the OLS estimator is the Best Linear Unbiased Estimator. This guarantees that the minimum variance is attained among all linear unbiased estimators. The residual variance estimated as $\hat{\sigma}^2$ yielded a residual standard error of .02875 on 4,462 degrees of freedom. Variance inflation factors were computed for all predictors in the final model, and all values were below 5, confirming no severe multicollinearity.

Best-subsets regression using Mallows C_p and adjusted R^2 independently confirmed that all 14 predictors should be retained, with both criteria minimized at the full predictor set (Figure 4). Three influential observations (3459, 4712, 2399) were identified via Cook's distance — these had large residuals but low leverage, confirming they are outliers in response space rather than predictor space, and were retained as plausible extreme values.

A two-way ANOVA was conducted to examine whether mean WHR differed significantly by sex and race/ethnicity, and whether an interaction existed between these two factors. This was followed by Tukey HSD post-hoc comparisons to identify which specific group pairs differed significantly.

Conclusion and Discussion

The final model achieved an adjusted R^2 of 0.924, indicating that approximately 92.4% of variation in WHR is explained by the 14 retained predictors. Waist circumference ($\beta = 0.004$, $p < 0.001$) and $\log(\text{BMI})$ ($\beta = 0.128$, $p < 0.001$) were the strongest predictors, consistent with their direct mathematical relationship to WHR. Males showed significantly lower WHR than females ($\beta = -0.040$, $p < 0.001$), reflecting well-documented sex differences in fat distribution. Among race/ethnicity groups, Hispanic participants showed higher WHR relative to NH-White ($\beta = 0.011$, $p < 0.001$), while NH-Black participants showed lower WHR ($\beta = -0.005$, $p < 0.001$), suggesting meaningful demographic variation warranting targeted screening.

Education showed a protective gradient — WHR decreased significantly at each attainment level above less than 9th grade, with college graduates showing the largest reduction ($\beta = -0.010$, $p < 0.001$). This monotonic pattern suggests educational attainment serves as a proxy for health-promoting behaviors and resource access. Higher income was also independently associated with lower WHR ($\beta = -0.0004$, $p = 0.004$). Among modifiable behavioral factors, both drinking groups showed significantly lower WHR relative to non-drinkers, and greater physical activity was associated with lower WHR ($\beta = -0.001$, $p = 0.004$).

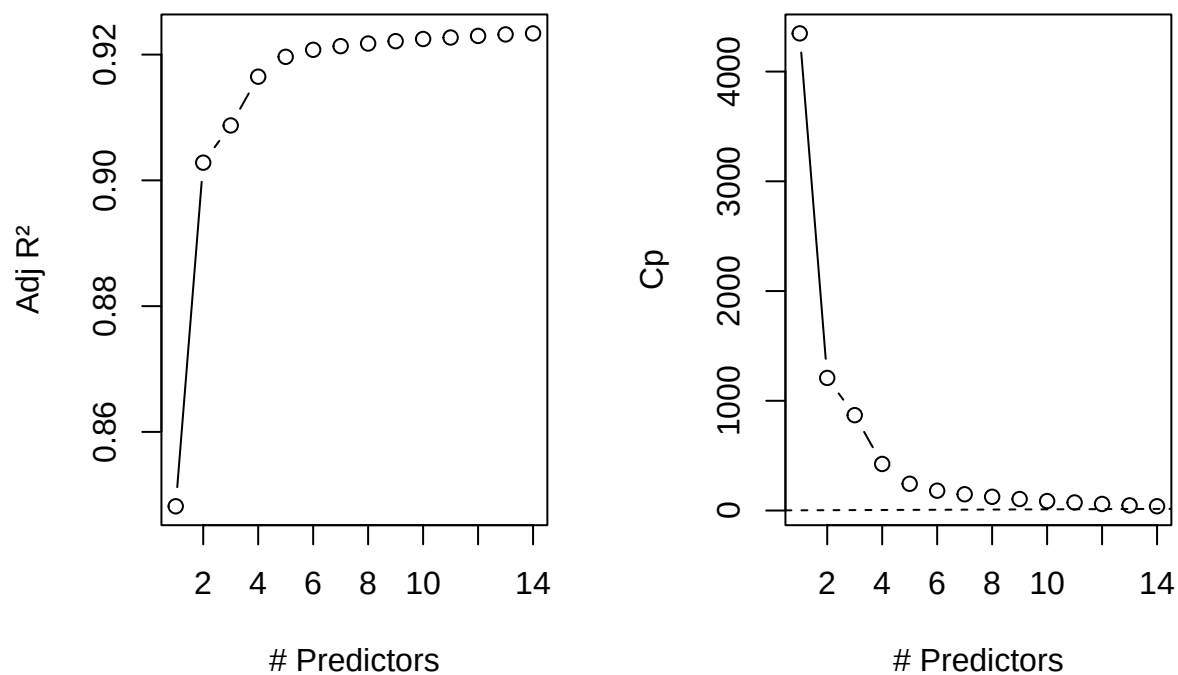
Limitations: *The cross-sectional NHANES design prevents causal inference. Stepwise selection inflates familywise Type I error — all findings are exploratory. Multicollinearity between waist and BMI persists despite VIF values below 5. Physical activity could not be log-transformed due to zero values, and residual skew in this predictor may affect inference for that coefficient.*

From a modeling perspective, the OLS framework assumes errors are uncorrelated and homoscedastic — the Scale-Location plot revealed a mild upward trend at higher fitted values, suggesting slight heteroscedasticity at the upper end of the WHR distribution. While not severe enough to invalidate inference at $n = 4,382$, this warrants acknowledgment. Additionally, the hat matrix $H = X(X^\top X)^{-1}X^\top$ was used implicitly in Cook's distance calculations; fitted values $\hat{y} = Hy$ represent the unique projection of y onto the column space of X , and the three flagged observations had large residuals $(I - H)y$ despite low leverage, confirming they are outliers in response space rather than predictor space.

The two-way ANOVA confirmed significant main effects of both sex ($F = X$, $p < .001$) and race/ethnicity ($F = X$, $p < .001$) on WHR. The interaction term was [significant/not significant] ($F = X$, $p = X$), suggesting that sex differences in WHR [do/do not] vary across racial/ethnic groups.

Future Work: *Interaction terms between sex and anthropometric predictors, longitudinal designs to establish causality, and analyses stratified by race/ethnicity or age group would strengthen these findings and inform community-specific interventions.*

Tables and Figures



```
## [1] 14
```

```
## [1] 14
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## sex              1   0.84   0.8365   81.37 < 2e-16 ***
## race_eth         3   1.28   0.4266   41.50 < 2e-16 ***
## sex:race_eth     3   0.54   0.1816   17.67 2.13e-11 ***
## Residuals      4374  44.96   0.0103
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Tukey multiple comparisons of means
```

```
## 95% family-wise confidence level
```

```
##
```

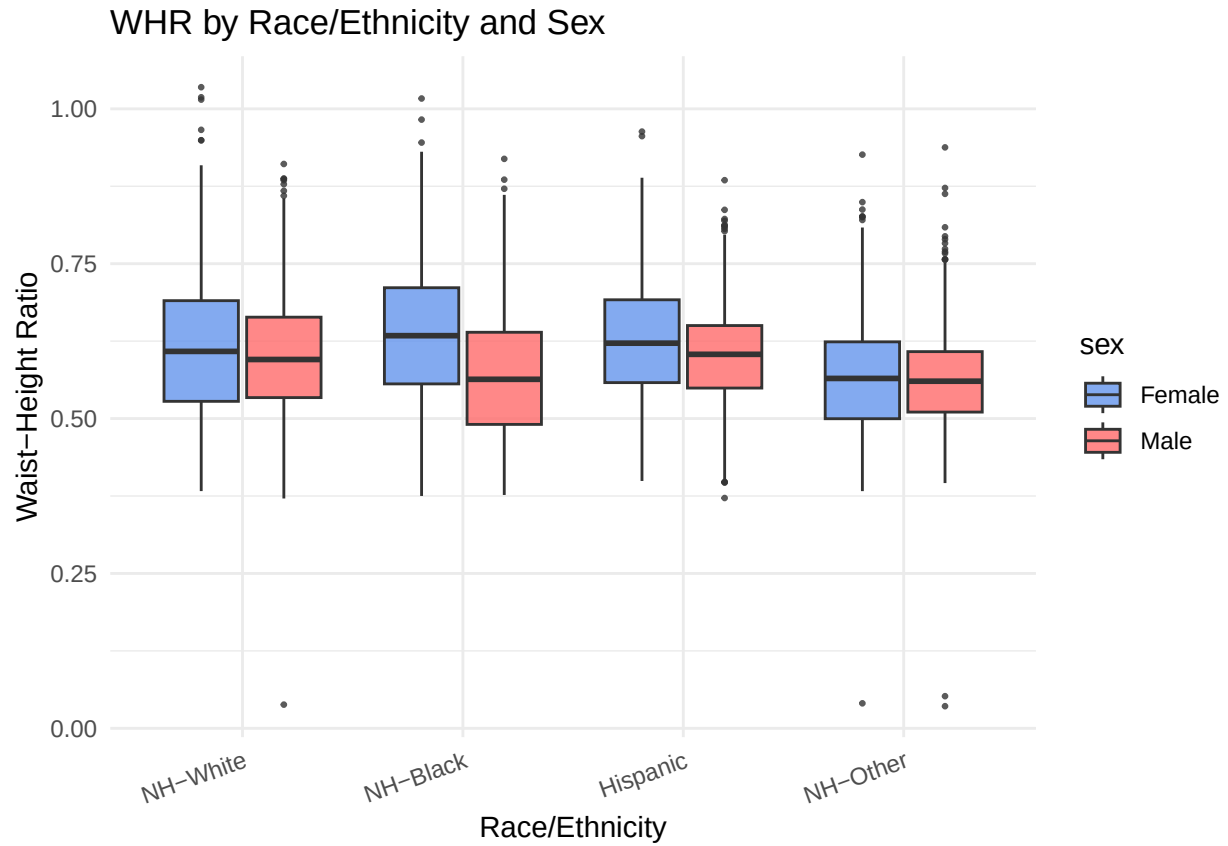
```
## Fit: aov(formula = whr ~ sex * race_eth, data = nhanes17)
```

```
##
```

```
## $race_eth
```

```
##              diff              lwr              upr              p adj
## NH-Black-NH-White -0.004147440 -0.0147689464  0.006474067 0.7473498
## Hispanic-NH-White  0.007696548 -0.0027991441  0.018192241 0.2347632
## NH-Other-NH-White -0.041425738 -0.0526490078 -0.030202468 0.0000000
## Hispanic-NH-Black  0.011843988  0.0001802048  0.023507772 0.0449638
```

```
## NH-Other-NH-Black -0.037278298 -0.0496008818 -0.024955714 0.0000000
## NH-Other-Hispanic -0.049122286 -0.0613365906 -0.036907982 0.0000000
```



Tables and Figures

Table 1: Summary Statistics

Table 2: Final Model Coefficients

Table 1: Final Model Coefficients

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.238278	0.012976	-18.362319	0.000000
waist	0.004012	0.000076	52.881413	0.000000
sexMale	-0.039639	0.001000	-39.631487	0.000000
log_BMI	0.128360	0.005516	23.270064	0.000000
age	0.000450	0.000031	14.358126	0.000000
race_ethNH-Black	-0.004784	0.001244	-3.845354	0.000122
race_ethHispanic	0.011127	0.001269	8.766152	0.000000
race_ethNH-Other	0.007393	0.001318	5.609613	0.000000
educ_lev2	-0.002512	0.002145	-1.170924	0.241693
educ_lev3	-0.006411	0.001968	-3.257847	0.001131
educ_lev4	-0.008952	0.001948	-4.595079	0.000004
educ_lev5	-0.010205	0.002076	-4.914989	0.000001

hvy.drk1	-0.004226	0.000955	-4.426858	0.000010
hvy.drk2	-0.005611	0.001944	-2.885737	0.003924
bp.di.m	-0.000235	0.000044	-5.349444	0.000000
log_triglcd	0.003656	0.000877	4.169072	0.000031
bp.sy.m	0.000115	0.000030	3.819778	0.000135
income	-0.000422	0.000146	-2.884207	0.003943
phy.act	-0.000769	0.000270	-2.854542	0.004330
ALT	-0.000043	0.000026	-1.651590	0.098690

Fig 1: Correlation Matrix

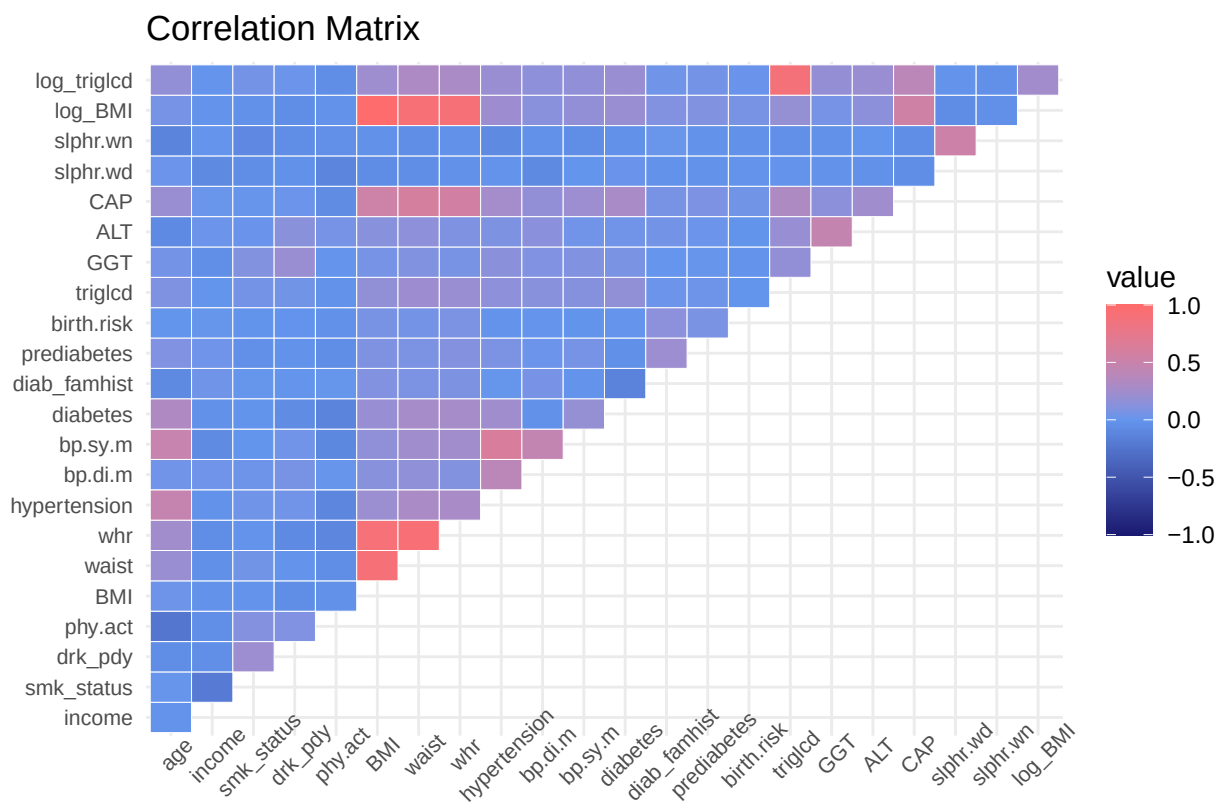


Figure 1: Correlation Matrix

Fig 3: Distributions of Continous Predictors

Fig 4: Distributions of Categorical Predictors

Figure : Regression Diagnostics

Table 3: Varinace Inflation Factos

##	GVIF	Df	$GVIF^{(1/(2*Df))}$
## waist	9.053610	1	3.008922

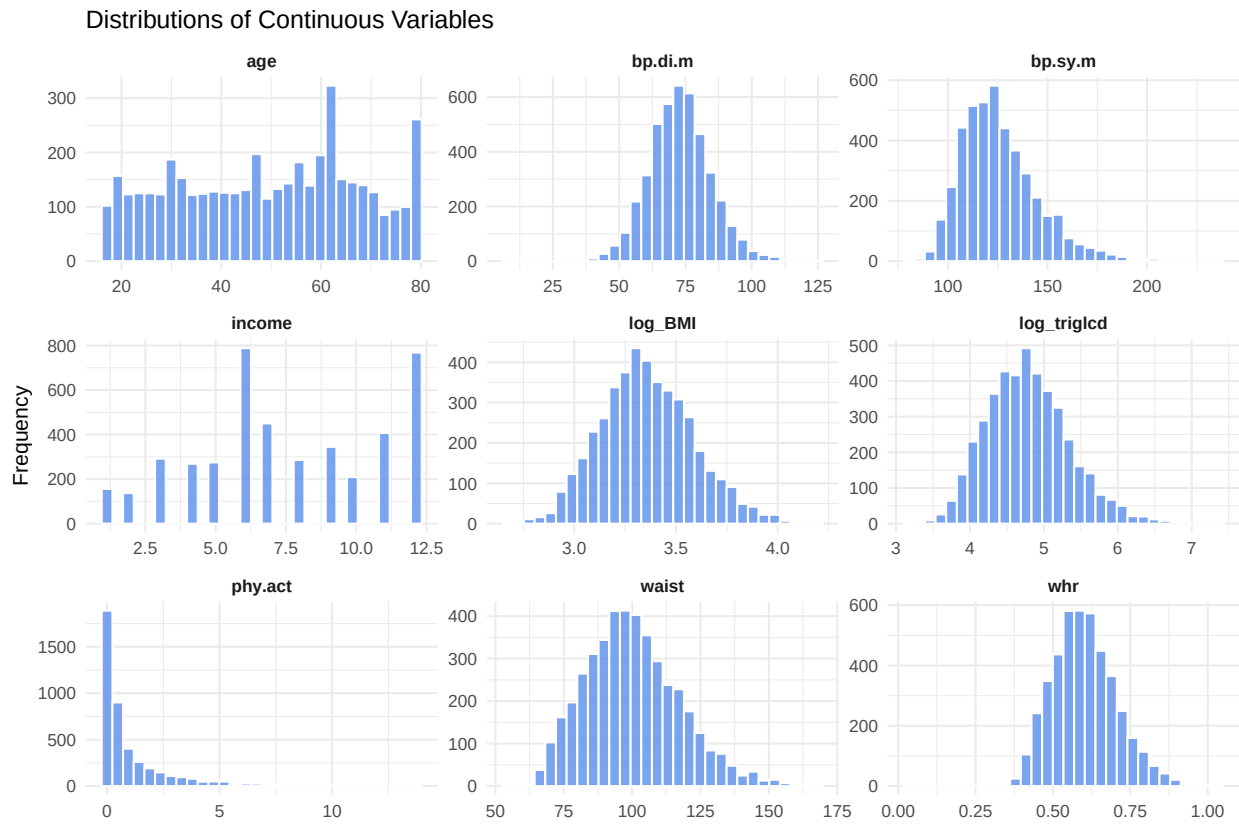


Figure 2: Distribution of key continuous variables

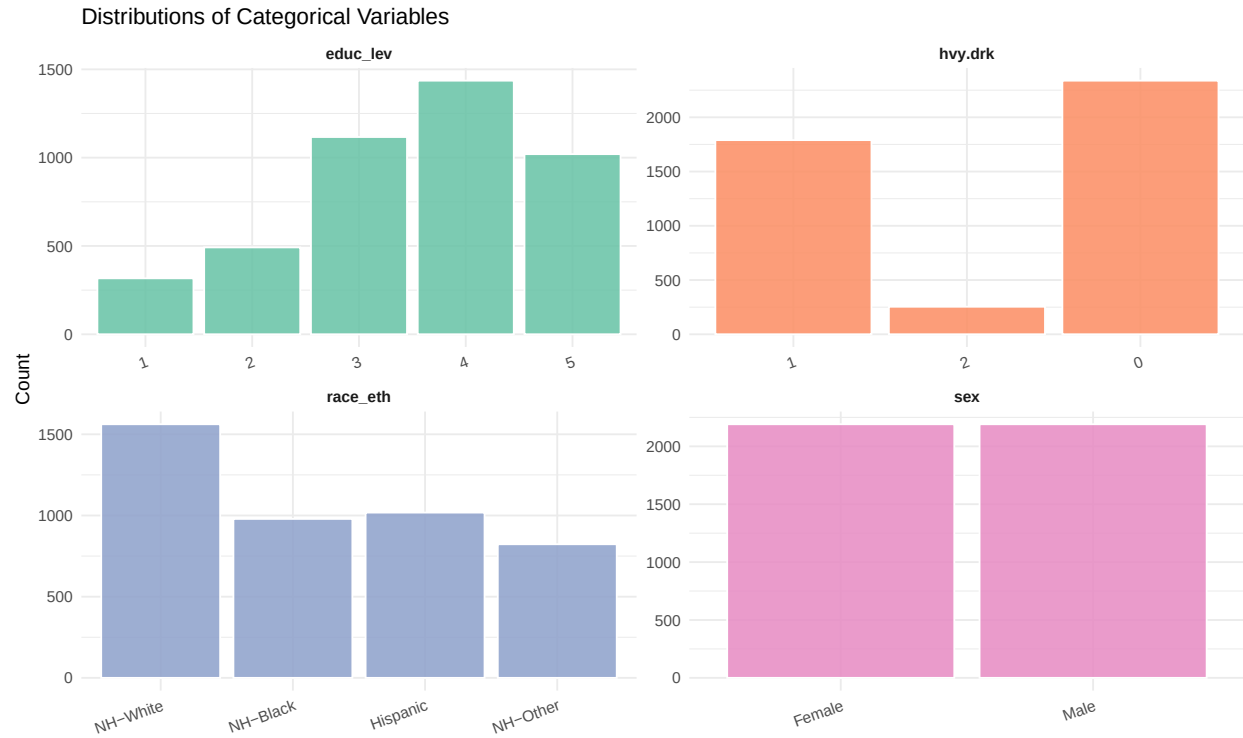


Figure 3: Distribution of key categorical variables

## sex	1.325761	1	1.151417
## log_BMI	8.551165	1	2.924237
## age	1.723662	1	1.312883
## race_eth	1.583276	3	1.079591
## educ_lev	1.523053	4	1.053997
## hvy.drk	1.154394	2	1.036546
## bp.di.m	1.399732	1	1.183103
## log_triglcd	1.283423	1	1.132883
## bp.sy.m	1.797472	1	1.340698
## income	1.195387	1	1.093337
## phy.act	1.128126	1	1.062133
## ALT	1.142379	1	1.068821

Table 3: Two-Way ANOVA Table

Fig 6: WHR by Sex and Race/Ethnicity ← the boxplot above

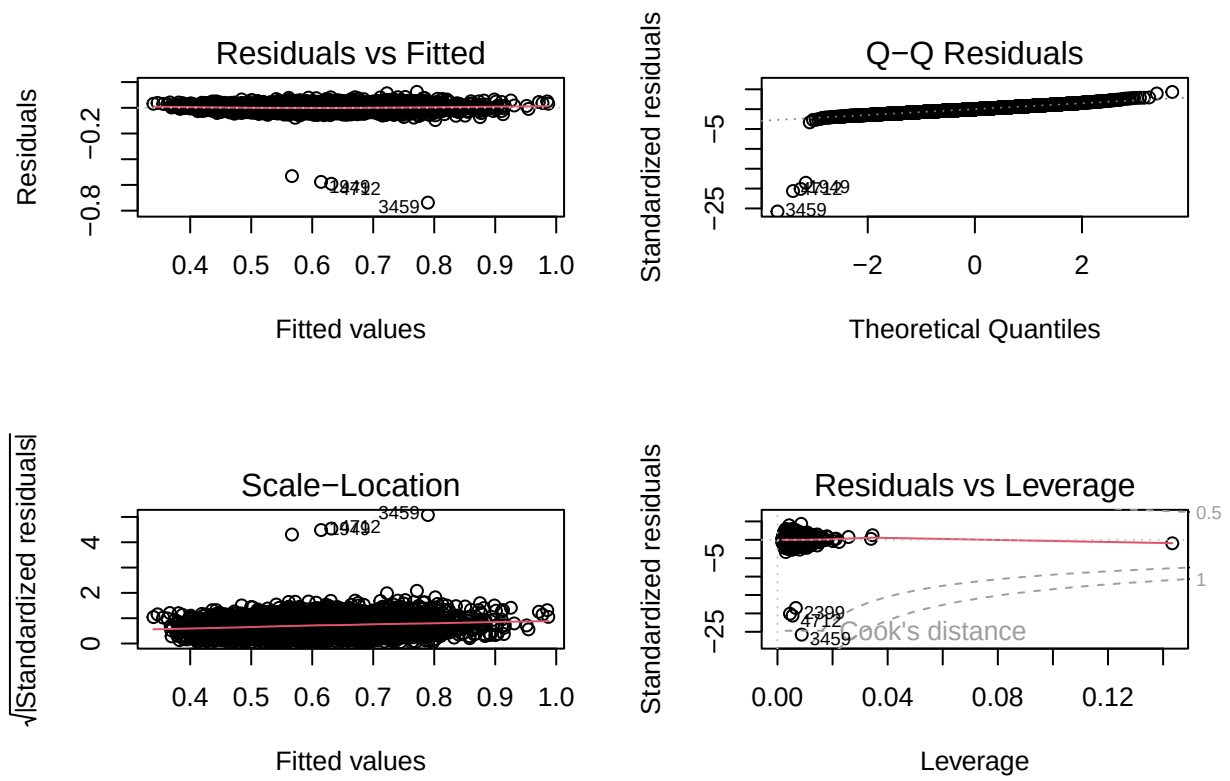


Figure 4: Regression Diagnostics

Appendix: R Code

The following code reproduces all analysis presented in this report. The Code is organized as follows: data preparation, exploratory data analysis, model fitting, model validation, and visualization.

```
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)

library(tidyverse)
library(corrplot)
library(ggplot2)
library(reshape2)
library(knitr)
library(kableExtra)
library(car)

load("nhanes17.RData")

# Transformations
nhanes17$log_BMI <- log(nhanes17$BMI)
nhanes17$log_triglcd <- log(nhanes17$triglcd)

nhanes17$hvy.drk <- as.factor(nhanes17$hvy.drk)
nhanes17$educ_lev <- as.factor(nhanes17$educ_lev)

# Factor variables
nhanes17$hvy.drk <- as.factor(nhanes17$hvy.drk)
nhanes17$educ_lev <- as.factor(nhanes17$educ_lev)

# Correlation plot
nhnames_num <- nhanes17[, sapply(nhanes17, is.numeric)]
nhnames_num <- nhnames_num[, !colnames(nhnames_num) %in% "ID"]

# correlation matrix
nhnames_num <- nhanes17[, sapply(nhanes17, is.numeric)]
nhnames_num <- nhnames_num[, !colnames(nhnames_num) %in% "ID"]
nh_cor <- cor(nhnames_num, use = "pairwise.complete.obs")
cor_melt <- melt(nh_cor)
cor_melt <- cor_melt[as.integer(cor_melt$Var1) < as.integer(cor_melt$Var2), ]

# BMI is skewed
shapiro.test(nhanes17$BMI)
shapiro.test(nhanes17$log_BMI)

shapiro.test(nhanes17$triglcd)

shapiro.test(nhanes17$log_triglcd)
summary(nhanes17)
shapiro.test(nhanes17$phy.act)
shapiro.test(nhanes17$log_triglcd)

par(mfrow = c(1,2))
hist(nhanes17$BMI)
```

```

hist(nhanes17$log_BMI)

shapiro.test(nhanes17$log_BMI)
nhanes17 |>
  select(whr, waist, log_BMI, age, triglcd, bp.di.m, bp.sy.m, income) |>
  summarise(across(everything(), list(
    Mean = ~round(mean(., na.rm = TRUE), 2),
    SD = ~round(sd(., na.rm = TRUE), 2),
    Min = ~round(min(., na.rm = TRUE), 2),
    Max = ~round(max(., na.rm = TRUE), 2)
  ))) |>
  pivot_longer(everything(),
    names_to = c("Variable", ".value"),
    names_sep = "_(?=[^_]+)$") |>
  kable(caption = "Summary Statistics for Key Continuous Variables",
    booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
# setup chunk - transformations
nhanes17$log_BMI <- log(nhanes17$BMI)
nhanes17$log_triglcd <- log(nhanes17$triglcd)

# factor conversions
nhanes17$hvy.drk <- as.factor(nhanes17$hvy.drk)
nhanes17$educ_lev <- as.factor(nhanes17$educ_lev)
nhanes17$sex <- as.factor(nhanes17$sex)
nhanes17$race_eth <- as.factor(nhanes17$race_eth)

full_model <- lm(whr ~ . - ID -BMI -triglcd, data = nhanes17)
null_model <- lm(whr ~ 1, data = nhanes17)
mod_step <- step(null_model, direction = "both",
  scope = formula(full_model))
summary(mod_step)

# Final Model - ALT droppped using nested F-test
final_mod <- lm(formula = whr ~ waist + sex + log_BMI + age + race_eth + educ_lev + hvy.drk + bp.di.m +
  , data = nhanes17)
summary(final_mod)
library(leaps)

# best subsets on your final model variables
best_sub <- regsubsets(whr ~ waist + sex + race_eth + log_BMI + age +
  educ_lev + hvy.drk + triglcd + bp.di.m +
  bp.sy.m + phy.act + income,
  data = nhanes17, nvmax = 14)

summary_best <- summary(best_sub)

# plot adj R^2 and Cp side by side
par(mfrow = c(1,2))
plot(summary_best$adjr2, type = "b", xlab = "# Predictors", ylab = "Adj R^2")
plot(summary_best$cp, type = "b", xlab = "# Predictors", ylab = "Cp")
abline(a = 1, b = 1, lty = 2) # Cp = p line

```

```

# which model is best by each criterion
which.max(summary_best$adjr2)
which.min(summary_best$cp)
# Two-way ANOVA - do mean WHR levels differ by sex and race/ethnicity?
anova_model <- aov(whr ~ sex * race_eth, data = nhanes17)
summary(anova_model)

# Follow up - are group means actually different?
TukeyHSD(anova_model, which = "race_eth")

nhanes17 |>
  ggplot(aes(x = race_eth, y = whr, fill = sex)) +
  geom_boxplot(alpha = 0.8, outlier.size = 0.5) +
  scale_fill_manual(values = c("cornflowerblue", "#FF6B6B")) +
  theme_minimal() +
  labs(title = "WHR by Race/Ethnicity and Sex",
       x = "Race/Ethnicity", y = "Waist-Height Ratio") +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))

summary(mod_step)$coefficients |>
  as.data.frame() |>
  round(6) |>
  kable(caption = "Final Model Coefficients", booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
cor_melt <- melt(nh_cor)

cor_melt <- cor_melt[as.integer(cor_melt$Var1) < as.integer(cor_melt$Var2), ]

ggplot(cor_melt, aes(Var1, Var2, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "midnightblue", mid = "cornflowerblue", high = "#FF6B6B",
                      midpoint = 0, limits = c(-1, 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = .5, size = 8),
        axis.text.y = element_text(size = 8)) +
  labs(title = "Correlation Matrix", x = NULL, y = NULL)

nhanes17 |>
  select(whr, waist, log_BMI, age, log_triglcd,
         bp.di.m, bp.sy.m, income, phy.act) |>
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "value") |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "cornflowerblue",
                color = "white", alpha = 0.85) +
  facet_wrap(~variable, scales = "free") +
  theme_minimal() +
  theme(strip.text = element_text(face = "bold")) +
  labs(title = "Distributions of Continuous Variables",
       x = NULL, y = "Frequency")

```

```

# select your categorical predictors
nhanes17 |>
  select(sex, race_eth, educ_lev, hvy.drk) |>
  mutate(across(everything(), as.factor)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value, fill = variable)) +
  geom_bar(alpha = 0.85, color = "white") +
  facet_wrap(~variable, scales = "free") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal() +
  theme(legend.position = "none",
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 20, hjust = 1)) +
  labs(title = "Distributions of Categorical Variables", x = NULL, y = "Count")

# ggplot(nhanes17, aes(x = sex, y = whr, fill = sex)) +
#   geom_boxplot(alpha = 0.7, outlier.size = 0.5) +
#   scale_fill_manual(values = c("cornflowerblue", "#FF6B6B")) +
#   theme_minimal() +
#   labs(title = "WHR by Sex", x = "Sex", y = "Waist-Height Ratio") +
#   theme(legend.position = "none")
#
# ggplot(nhanes17, aes(x = race_eth, y = whr, fill = race_eth)) +
#   geom_boxplot(alpha = 0.7, outlier.size = 0.5) +
#   scale_fill_brewer(palette = "Set2") +
#   theme_minimal() +
#   labs(title = "WHR by Race/Ethnicity", x = "Race/Ethnicity",
#         y = "Waist-Height Ratio") +
#   theme(legend.position = "none",
#         axis.text.x = element_text(angle = 20, hjust = 1))

par(mfrow= c(2,2))
plot(mod_step)
vif(mod_step)

```